

Part B: Take-Home Test

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Moderator: Mr RQ Odendaal

Release Date: 08 June 2021 at 14:45
Due Date: 08 June 2021 at 16:45
Grace Period: 16:45 to 17:15

Please read and understand the following.

1. All test and examination regulations of the University of Pretoria apply.
2. This is an OPEN BOOK test. You may use any source available to you, like your textbook and lecture notes, to assist in answering the questions. However, you **MAY NOT** receive (or provide) help in any way whatsoever from (or to) any other person. This includes the lecturer, tutor, a friend and family members.
3. Answer the questions **in your own handwriting** on separate paper, lined or unlined.
Take quality portrait orientation scans of your answer sheets and save them as PDF's. Use apps like **CamScanner** to scan and to convert photographs to PDF.
Submit your answer sheets, via ClickUP, in the "Semester Test 2 (Part B Take-Home Assignment) Submission" in the "**Semester and Online Tests Folder**" under the "**TESTS & EXAMS**" headline.
Submit before 17:15.
There will be a queue during submission. You may not get your submission receipt immediately! To accommodate this, we give you a grace period of 30 minutes only, between 16:45 and 17:15 to make your submission. Record any problems submitting at <https://forms.gle/VaQWq3PLPTZAMDRH6>
4. **YOU HAVE BEEN WARNED: Late submissions will not be accepted!**
5. Number and label your submissions as in the question paper. Incorrectly-numbered and incorrectly-labelled answers may not be marked.
6. Provide neatly labelled sketches, including co-ordinate systems, free-body diagrams, final answers with correct numbers of significant figures, units, etc. Failure to do so will result in marks being deducted.
7. Your answer needs to be written in a systematic and logical fashion. Map out the steps you will take to reach your goal i.e. solve the problem in your mind first! You must show that you understand and can use the relevant physics. Only substitute numerical values in the final step of calculations! A correct answer does not necessarily mean that you will receive full marks!
8. Units must be included when numbers are substituted; else, marks will be deducted.
9. You will need to look up constants and typical values for parameters necessary in your calculations. Use your textbook or other credible sources.
10. If any misconduct with respect to this assessment is suspected, you will receive a mark of zero and will have to attend a formal disciplinary hearing, which may result in your expulsion from the University of Pretoria.

DECLARATION OF ORIGINALITY

UNIVERSITY OF PRETORIA - Department of Physics

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I understand what plagiarism is and am aware of the University's policy in this regard. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

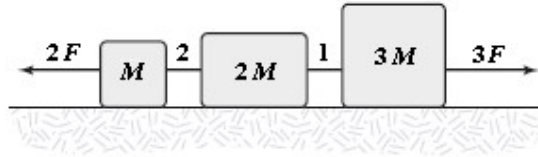
This statement is a reminder of the Code of Conduct that you have already signed.

TOTAL POINTS (Part B): 43

Question 1 [15]

1.1 Show that the total force needed to drag a box at constant speed v across a horizontal surface with coefficient of kinetic friction μ_k is least when the force is applied at an angle θ with respect to the $+x$ -axis (or the horizontal surface) such that $\tan \theta = \mu_k$. (10)

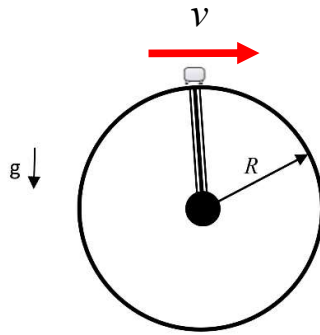
1.2 All the masses in the accompanying figure slide on a horizontal frictionless surface. If $F = 12$ N, determine T_1 , the tension in string 1. Treat the strings as being stretch-less and of negligible mass. (5)



Question 2 [16]

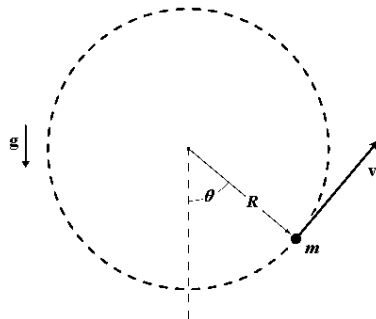
2.1 A car enters a level, unbanked semi-circular hairpin bend with a radius of 300 m at a speed of 120 km/h. The coefficient of friction between the tyres and the road is $\mu = 0.250$. If the car maintains the constant speed of 120 km/h, show that the likely fate for the car will be to veer toward the outside of the circle. (6)

2.2 An amusement ride consists of a car moving in a vertical circle on the end of a rigid boom. The radius of the circle is 10 m. The combined weight of the car and riders is 5.0 kN. At the top of the circle the car has a tangential speed of 5.0 m/s which is not changing at that instant. What is the force of the boom on the car at the top of the circle? (5)



2.3 A 0.30-kg mass attached to the end of a string swings in a vertical circle ($R = 1.6$ m), as shown. At an instant when $\theta = 50^\circ$, the tension in the string is 8.0 N. What is the magnitude of the resultant force on the mass at this instant?

CAUTION: be careful about the type of circular motion that the mass is undergoing. (5)

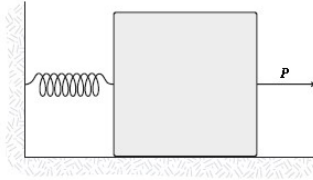


Question 3 [12]

3.1 A single conservative force $F_x = (6.0x - 12)$ N (x is in m) acts on a particle moving along the x axis. The potential energy associated with this force is assigned a value of +20 J at $x = 0$. What is the potential energy at $x = 3.0$ m? (5)

3.2(a) A block of mass m on a horizontal frictionless surface is attached to a light spring with force constant k . The block is initially at rest at its natural (equilibrium) position when a force P acting parallel to the surface, is applied to the block as shown. What is the speed of the block when it is at position x from its equilibrium position? (5)

3.2(b) If the block has a mass of 12 kg, the spring constant has a value of 0.80 kN/m when the block is initially at rest at the equilibrium position when a force of magnitude $P = 80$ N acting parallel to the surface is applied to the block. What is the speed of the block when it is 13 cm from the equilibrium position? (2)



***** E N D S *****